Midterm

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Problem 1

$$g_{\mu\nu} = \begin{pmatrix} -\left(1-\frac{2M}{\sqrt{r^2+b^2}}\right) & 0 & 0 & 0 \\ 0 & \left(1-\frac{2M}{\sqrt{r^2+b^2}}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2+b^2 & 0 \\ 0 & 0 & 0 & (r^2+b^2)\sin^2\theta \end{pmatrix}$$

$$g^{\mu\nu} = \begin{pmatrix} -\left(1-\frac{2M}{\sqrt{r^2+b^2}}\right)^{-1} & 0 & 0 & 0 \\ 0 & \left(1-\frac{2M}{\sqrt{r^2+b^2}}\right) & 0 & 0 \\ 0 & 0 & \frac{1}{r^2+b^2} & 0 \\ 0 & 0 & 0 & \frac{1}{(r^2+b^2)\sin^2\theta} \end{pmatrix}$$

Problem 2

$$
\begin{aligned}
\Gamma^t_{tr} &= \frac{Mr}{(r^2+b^2)\left(\sqrt{r^2+b^2}-2M\right)} \\
\Gamma^r_{tt} &= \frac{Mr\left(\sqrt{r^2+b^2}-2M\right)}{(r^2+b^2)^2} \\
\Gamma^r_{rr} &= -\frac{Mr}{(r^2+b^2)\left(\sqrt{r^2+b^2}-2M\right)} \\
\Gamma^r_{\theta\theta} &= r\left(\frac{2M}{\sqrt{r^2+b^2}}-1\right) \\
\Gamma^r_{\phi\phi} &= r\left(\frac{2M}{\sqrt{r^2+b^2}}-1\right)\sin^2\theta \\
\Gamma^\theta_{\theta r} &= \frac{r}{r^2+b^2} \\
\Gamma^\theta_{\phi\phi} &= -\sin\theta\cos\theta \\
\Gamma^\phi_{\phi r} &= \frac{r}{r^2+b^2} \\
\Gamma^\phi_{\phi\theta} &= \cot\theta
\end{aligned}
$$

Problem 3

$$
\begin{aligned}
\ddot{t} &= -\frac{2Mr\dot{t}\dot{r}}{(r^2+b^2)\left(\sqrt{r^2+b^2}-2M\right)} \\
\ddot{r} &= \frac{r}{(r^2+b^2)^2\left(\sqrt{r^2+b^2}-2M\right)}\left[\left(\sqrt{r^2+b^2}-2M\right)^2\left((r^2+b^2)^{\frac{3}{2}}\left(\sin^2\theta\dot{\phi}^2+\dot{\theta}^2\right)-M\dot{t}^2\right)+M(r^2+b^2)\dot{r}^2\right] \\
\ddot{\theta} &= \sin\theta\cos\theta\dot{\phi}^2-\frac{2r\dot{\theta}\dot{r}}{r^2+b^2} \\
\ddot{\phi} &= -2\dot{\phi}\left(\cot\theta\dot{\theta}+\frac{r\dot{r}}{r^2+b^2}\right)
\end{aligned}
$$

Problem 4

The metric is independent of t and ϕ, so we have two conserved quantities:

$$
\begin{aligned}
e &\coloneqq -\xi_t{}^\mu u_\mu = \left(1-\frac{2m}{\sqrt{r^2+b^2}}\right)\dot{t} \\
l &\coloneqq -\xi_\phi{}^\mu u_\mu = -(r^2+b^2)\sin^2\theta\dot{\phi}
\end{aligned}
$$

The ϕ independence also implies that orbits lie in a plane, so we can consider an orbit in the $\theta = \pi/2$ plane, with $\mathrm{d}\theta = 0$.

$$
\begin{aligned}
u^2 = -1 &= -\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)\dot{t}^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1}\dot{r}^2 + (r^2+b^2)\dot{\phi}^2 \\
&= -\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1} e^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1}\dot{r}^2 + \frac{l^2}{r^2+b^2} \\
\implies \qquad e^2 &= \dot{r}^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)\left(1 + \frac{l^2}{r^2+b^2}\right) \\
\implies \qquad \frac{1}{2}(e^2-1) &= \frac{1}{2}\dot{r}^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)\left(1 + \frac{l^2}{r^2+b^2}\right) - 1 \\
\implies \qquad \mathcal{E} &= \frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) \qquad (1)
\end{aligned}
$$

where

$$
\begin{aligned}
V_{\text{eff}}(r) &\coloneqq \frac{1}{2}\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)\left(1 + \frac{l^2}{r^2+b^2}\right) - 1 \\
&= -\frac{M}{\sqrt{r^2+b^2}} + \frac{l^2}{2(r^2+b^2)} - \frac{Ml^2}{(r^2+b^2)^{\frac{3}{2}}} \qquad (2)
\end{aligned}
$$

Problem 5

No, for $M = 0$ bound orbital solutions do not exist. In this case, the effective potential is of the form

$$V_{\text{eff}}(r)\big|_{M\to 0} = \frac{l^2}{2(r^2+b^2)},$$

which has a single maximum at $r = 0$ and no local minima, as illustrated in Figure 1.

Figure 1: An example plot of V_{eff} with $M = 0$.

Problem 6

$$
\begin{aligned}
\frac{\mathrm{d}r}{\mathrm{d}\tau} &= \frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}\tau} \\
&= \frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{-l}{r^2+b^2} \\
\implies \qquad \mathcal{E} &= \frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{-l}{r^2+b^2}\right)^2 + V_{\text{eff}}(r) \\
\implies \qquad \frac{\mathrm{d}r}{\mathrm{d}\phi} &= \pm\frac{r^2+b^2}{l}\sqrt{2(\mathcal{E}-V_{\text{eff}})}
\end{aligned}
$$

Problem 7

$$
V_{\text{eff}}(r) = \frac{-1}{\sqrt{r^2+1}} + \frac{25}{2(r^2+1)} - \frac{25}{(r^2+1)^{\frac{3}{2}}}
$$

Minimum occurs ar $r \approx 21.5$ Bound orbits exist.

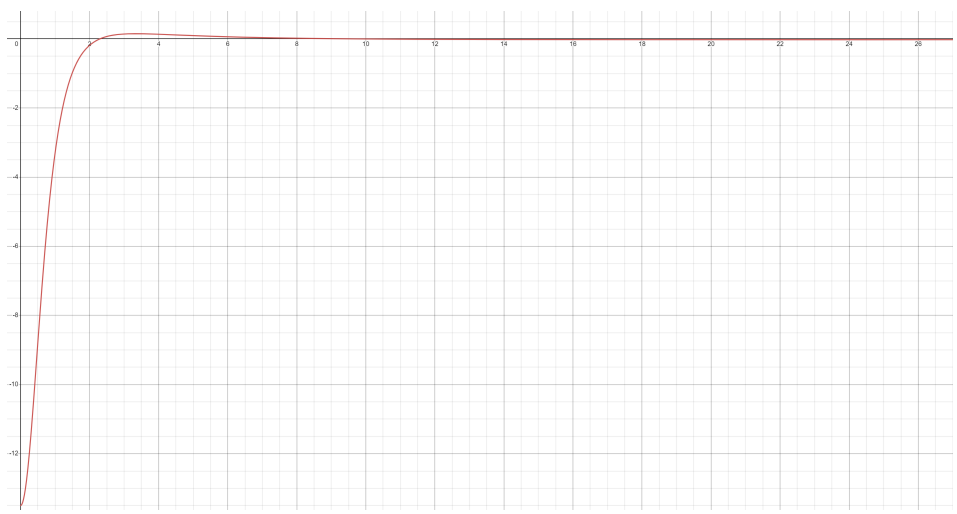

Figure 2: $V_{\text{eff}}(r)$ with $M = 1$, $b = 1$, $l = 5$.

Problem 8

$$
V_{\text{eff}}(r) = \frac{-1}{\sqrt{r^2+9}} + \frac{4}{2(r^2+9)} - \frac{4}{(r^2+9)^{\frac{3}{2}}}
$$

$V_{\text{eff}} = \mathcal{E}$ when $r \approx 17.8$, so orbits are bound between $r = 0$ and this maximum.

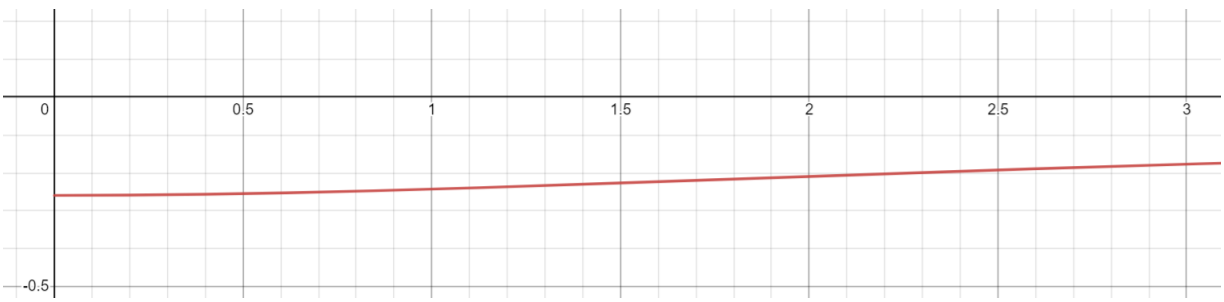

Figure 3: $V_{\text{eff}}(r)$ with $M = 1$, $b = 3$, $l = 2$.